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EDUCATION

RUTGERS SCHOOL OF ARTS AND SCIENCES, HONORS PROGRAM

B.S in Computer Science, B.S in Data Science(Economics Track)

NEW BRUNSWICK, NJ

Class of 2027

• **Relevant Coursework:** Intro to Business Analytics, Statistics I, Data 101, Intro to Computer Science, Data Structures, Discrete Structures I, Intro Linear Algebra, Computer Architecture, Discrete Structures II, Data Management, Design and Analysis of Algorithms, Intro to Microeconomics, Intro Macroeconomics

HIGHLAND PARK HIGH SCHOOL, NATIONAL HONORS SOCIETY

GPA: 4.56/4.0

HIGHLAND PARK, NJ

Class of 2023

PROJECTS

•BCG Data Science Job Simulation on Forage - May 2025

Tools: Python, Pandas, NumPy, SciKit Learn

Completed a customer churn analysis simulation for XYZ Analytics, demonstrating advanced data analytics skills, identifying essential client data and outlining a strategic investigation approach. Conducted efficient data analysis using Python, including Pandas and NumPy. Employed data visualization techniques for insightful trend interpretation. Completed the engineering and optimization of a random forest model, achieving an 85% accuracy rate in predicting customer churn. Completed a concise executive summary for the Associate Director, delivering actionable insights for informed decision-making based on the analysis.

•Unsupervised Quantitative Trading Model

Tools: Python, Pandas, Numpy, Scikit-Learn, Matplotlib, Pandas TA

This quantitative trading strategy utilizes KMeans and unsupervised machine learning techniques to enhance returns in stock trading. This project makes use of feature engineering with technical indicators, returns analysis, and Fama-French factors based on historical to current spanning S&P trading data. It then applies K-Means Clustering with a customized centroid determined from RSI factors. Optimized portfolio with Efficient Frontier analysis to achieve a Sharpe ratio approximately 20% higher than market benchmark.

•Blockchain-Based Fraud Detection System (In Progress)

Tools: Python, Pandas, Scikit-Learn, PyTorch, Web3.py, PostgreSQL, AWS, FastAPI, Flask, React Native, Docker

Developing a scalable fraud detection system for Ethereum transactions using machine learning. Conducting extensive exploratory data analysis to identify fraudulent patterns. Implementing deep learning models (PyTorch) for anomaly detection. Utilizing Web3.py for blockchain transaction monitoring and PostgreSQL for secure data storage. Deploying APIs with FastAPI and Flask for real-time fraud detection. Integrating AWS cloud services for model training and scalability. Designing an interactive mobile front-end in React Native for financial analysts to monitor risk in live transactions.

•**SkinSight:** Hackathon Project in a team of 4. Deep Learning Model utilizing Python with Pycharm IDE to detect common skin problems. Took into account computer vision detecting how different skin issues present themselves on several different skin tones. Utilized an inclusive database and trained model to 75% accuracy.

PROFESSIONAL EXPERIENCE/LEADERSHIP EXPERIENCE

Dr. Jung's Lab of Social Work

Undergraduate Research Assistant

NEW BRUNSWICK, NJ

May 2025 - Present

•Work with QGIS and Geospatial interfaces and utilize machine learning and deep learning algorithms to identify various poverty and climate change related statistics and patterns to assist with global public policy and relief programs. Assist with literature review, organizing meetings, and shadow alongside professors, graduate students, and PhD candidates.

RUTGERS MOBILE APPLICATION DEVELOPMENT(RUMAD) INCUBATOR

Software Engineer

NEW BRUNSWICK, NJ

Feb 2025 - Apr 2025

•Selected for a competitive incubator program to develop and deploy scalable mobile applications. Collaborated with a multidisciplinary team to develop a dynamic scheduling app for university students, leveraging LLMs to process chats and update schedules in real time. Users can upload images of their schedule, which the system edits dynamically. Features include friend integration via contacts or NetID for easier hangout planning. Built with React.js for the frontend, Flask for the backend, and SQLite/MongoDB for data management. Gemini API LLMs deep learning power the chatbot for intelligent scheduling.

RUTGERS UNIVERSITY

Learning Assistant for Data Structures CS112 and Introduction to Computer Science CS111

NEW BRUNSWICK, NJ

Aug 2024 - Present

• Responsible for Reviewing Lecture Content in Recitations. Help Students Grasp a Strong Understanding on Data Structures and Algorithms and their applications. Clarify student doubts, facilitate in class group work and problem solving sessions, and prepare students for exams. Occasionally supported faculty during lecture. Commended for excellence in presentation.

PROGRAMS/ACTIVITIES/ LEADERSHIP

• **Rutgers Mobile Application Development Events Committee Chair:** Plan club events with companies and panel speakers along with casual resume review and networking events for Rutgers Students

•**Highland Park High School Robotics Club President/Team Captain + Treasurer (Former)**

• **Goldman Sachs Virtual Insight Series, Bank of New York Mellon Freshman Jumpstart Program, Bank of New York Mellon Sophomore Summit**

SKILLS

Programming Languages: Python, Java, R, SQL, C, JavaScript

Web Development: React.js, React Native, Flask, Spring Boot, RESTful APIs

Databases: PostgreSQL, MongoDB

Cloud: AWS, Azure

Tools & Frameworks: Git, PyTorch, TensorFlow, Keras, Scikit-Learn, Pandas, Spark

Other: Jupyter, Tableau, Figma, HTML, CSS

Soft Skills: Strong written and oral communication skills